

SecOps AI Ticket Intelligence Platform

Project Plan • Semester 1 Prototype (MVP) with Semester 2 DevSecOps Deployment Goals

Version 1.0

1. Project Overview

This project delivers a software tool that connects to AutoTask (Datto) and provides a SecOps-focused, AI-assisted workflow for triaging and organising security tickets. AutoTask remains the 'source of truth'. The new tool consumes ticket data (read-only), enriches it with AI-generated category and priority, and presents it in a simplified dashboard.

1.1 Background and Problem Statement

The SecOps team in Global 4 receive a high volume of tickets generated automatically (e.g., Datto RMM alerts). In AutoTask, these tickets can appear as a flat list and may include many unused attributes, making it difficult to rapidly identify the most urgent incidents. Manual triage increases response time and cognitive load.

1.2 Objectives

Outlined below are the key objectives we must achieve, as approved and requested by SecOps:

- Provide a user-centric interface that simplifies SecOps ticket triage.
- Automatically categorise tickets into operational buckets (e.g., backup failed, patch failed) using AI/NLP.
- Generate prioritisation scores and short explanations to support analyst decision-making.
- Maintain AutoTask as the authoritative record (read-only integration).
- Deliver a working MVP on localhost using synthetic data, with a clear migration path to the real AutoTask REST API in Semester 2.

1.3 Scope

1.3.1 In scope (Semester 1 MVP):

- Synthetic ticket dataset closely matching SecOps AutoTask ticket attributes and Datto RMM-style text.
- Backend API that serves tickets and AI enrichment to the frontend.
- AI module integration point and basic working classifier (to be replaced/extended by the AI specialist).
- Frontend dashboard that displays categories and prioritised ticket lists.

1.3.2 Out of scope (Semester 1 MVP):

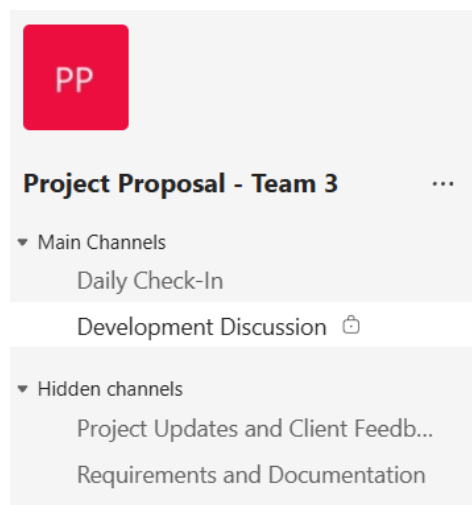
- Direct write-back or modification of AutoTask tickets.
- Use of real customer/company data.
- Full multi-tenant authentication/authorisation and auditing (planned for Semester 2).

2. Stakeholders and Roles

Role	Owner	Responsibilities
Project Lead / Backend Lead	Liam	Architecture, backend API, provider design (fake → real AutoTask), integration planning, project governance.
AI Specialist	Toby	Ticket NLP/weighting model, category assignment, prioritisation logic, explainability output.
Frontend/UI Lead	Khyati	Dashboard UX, category views, ticket detail view, API integration, usability improvements.

3. Communication Plan

In order to adhere to professional working standards, our group used and will continue to use a designated Teams group. The project lead has taken the liberty to create the appropriate channels within this, to ensure that conversations are grouped appropriately, and that communication stays structured and unambiguous. Below is an example of this:



The majority of conversations within the group take place in the Development Discussion channel, as our communications are largely based on development, however Daily Check-In is used for general questions. Project Updates and Client Feedback include SecOps members directly in order to streamline our communication with them. We understand they must receive an influx of emails

naturally, so we opted for a channel with Management and SecOps, and a private group chat just with SecOps:

▼ Members and guests (7)

Name	Title	Location	Tags ⓘ	Role	
 Jonathan White			Managers +1	Guest	×
 Peter Halasz			Managers +1	Guest	×
 Claire Berry			Managers +1	Guest	×
 Robbie Down			SecOps +1	Guest	×
 Kieran Walker			SecOps +1	Guest	×
 Phil Gent			SecOps +1	Guest	×
 Liam Catlin				Guest	×

People (7)

 (s) Khyati Bhatnagar
You

 (s) Liam Catlin

 (s) Toby Wood 1

 Kieran Walker (Guest)

 Phil Gent (Guest)

 Robbie Down (Guest)

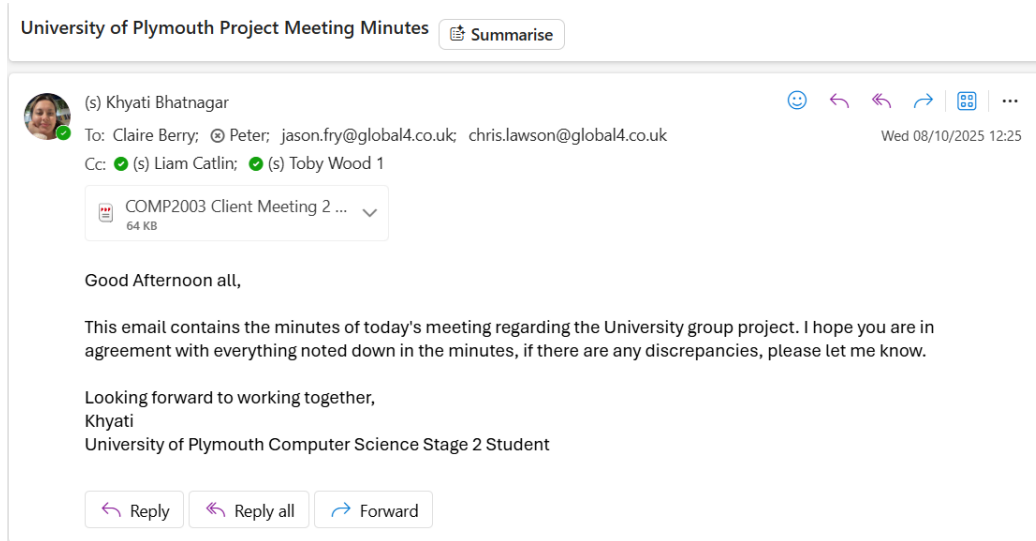
 Vassilis Cutsuridis

 Add people

 Add agents and bots

 Leave

Before the inclusion of Global 4 in Teams, we liaised with them via emails:



See 18.1 Appendices for minutes taken in Client Meetings.

4. Research

Research was conducted across the three core components of the solution to ensure that our approach was both technically sound and aligned with industry expectations. This included investigating how best to integrate the backend with AutoTask, understanding the requirements and constraints of developing an AI-driven ticket organiser, and examining the specific benefits such a system could provide to SecOps in terms of efficiency, clarity, and workload reduction. Alongside this, we carried out research into established UX principles and current design standards within the industry to ensure that the final interface would appear professional, meet user expectations, and provide the necessary features in a clear and intuitive manner.

The insights gained from researching the benefits of AI integration and automation enabled us to more effectively present and justify our proposed solution to Global 4 during the initial meetings.

See 18.2 and 18.3 Appendices for research documents.

5. Approval Process

The approval process consisted of firstly conveying the context of completing a group project for Global 4 and what they would like addressed; this was done in an initial meeting with the Management team. We were made aware of NDAs we would need to sign and send over, and we promptly did so. Secondly, our team presented to SecOps the proposed solution, separated into 2 key aspects—the backend and the frontend. Delivering the proposed solution, gave them an opportunity to clarify on expectations and highlight the satisfied components. Our team conveyed the importance of receiving an official list of requirements for the development process. Though there were some delays in receiving this, owing to the hectic schedule of SecOps, we were able to obtain this in our chat:

Kieran Walker (Guest) 18/11/2025 10:52

KW

We're looking at a software tool that plugs into AutoTask and helps us work smarter with alerts and tickets. Main things it should do:

1. Filter and automate alerts
 - Spot duplicate alerts and flag them
 - Match alerts to the right companies
 - Prioritise alerts and fill in UDF fields automatically
2. Easy-to-use interface
 - A GUI that lets you quickly sort, prioritise, and understand incoming tickets
 - Makes it easier to work through tasks quickly
3. Integration with AutoTask
 - Use the AutoTask REST API so AutoTask stays the "source of truth"
 - Extra features through the tool but without messing with existing AutoTask data

Basically, we want something that makes handling alerts and tickets faster and clearer, while keeping AutoTask as the main record

Phil is happy with this as well 😊

Any questions please let us know !

❤️ 2 🗨️

6. Budget

This project incurs no financial costs, as all required technologies are open-source. This choice also supports sustainability by reducing reliance on paid, proprietary tools and ensuring long-term accessibility.

7. User Requirements

7.1 Functional Requirements

The proposed solution must meet the following requirements:

- Connect/branch off AutoTask and present SecOps tickets in a clearer workflow without altering AutoTask data.
- Filter and automate alert handling for tickets **specific for SecOps** (queue: MS - SecOps).
- Detect duplicate or near-duplicate alerts and flag them for analysts.
- Match alerts to the correct companies and surface essential metadata at a glance.
- Prioritise alerts using AI and prepare for automated filling of relevant fields (internal fields in our tool; AutoTask UDF write-back is not in MVP).

7.2 UI/UX Requirements

SecOps requested that we do not redesign AutoTask completely; one of the reasons being that it would be difficult to take time off purely for learning the new system. As a result, one of the requests

made was that we create a UI that is essentially an improved version of AutoTask, which is much simpler to use and understand. Below are the requirements for the UI in more depth:

- Easy-to-use GUI that enables sorting, prioritisation, and rapid understanding of new tickets.
- Homepage showing urgent tickets due soon/immediately.
- Active tickets page grouped by AI categories with condensed, relevant attributes.
- Ticket detail page with side panel of all attributes and main panel for description and progress log.
- Closed tickets page with similar layout to active tickets.

7.3 Non-Functional Requirements

Presented below is a set of non-functional requirements which, although do not directly add new functionality, are still essential to the system's overall quality and long-term viability:

- Security: do not handle real production data in MVP; use synthetic data only.
- Maintainability: clear separation between data source (provider), AI logic, and API presentation.
- Scalability: design must support multiple SecOps users and multiple queues in the long term.
- Performance: ticket listing and AI enrichment should be fast enough for interactive dashboards.
- Future non-functional/feature requirements (Semester 2):
- Integrated chatbot (LLM) to answer questions and summarise ticket trends.
- Personal task organiser per staff member, settings/customisation, profile pages.
- Improved data management and auditing for operational use.

8. UI/UX Design

The UI is designed for operational clarity rather than feature density. This design is achieved through highlighting urgency using category grouping and AI priority scores, as well as displaying key relevant information at first glance, which reduces noise.

8.1 Pages

The User Interface consists of the following pages:

- Homepage: urgent and due-soon tickets, key workload indicators.
- Active Tickets: category buckets (Email blocked/held, Backup failed, Backup suspended, Hardware offline, Patching vulnerabilities, Patch failed).
- Ticket Detail View: attributes panel + description/progress log.
- Closed Tickets: archived view and quick search/filter.

See 18.4 Appendices for wireframes visually detailing this.

8.2 Interaction Model

- Analyst selects category → sees tickets sorted/grouped by priority.
- Click a ticket → opens detail view for investigation and notes (internal notes in our tool for MVP).
- Filters: status, priority, company, due date, and category.

9. AI Design

9.1 AI Purpose

The AI is used to support SecOps triage by extracting the intent of a ticket from a Datto RMM-style title and descriptions and organises them by the category the ticket should fall under, the company that raised the ticket, and give it a dynamic priority score based on the category and a number of other factors. The AI does not replace analysts: it gives triage suggestions to better inform the SecOps team.

9.2 Input

The input for tickets is the entire original ticket metadata – this ensures that all data is being analysed and looked at. It also takes an AI prompt as a static input when generating the AI description.

9.3 Outputs

New AI ticket, outputted in a JSON format.

This includes the category, the priority score and the company

This formats the ticket metadata into the critical information, with a focus on allowing uniformity throughout all tickets to make it easier to understand, at a glance, what a ticket is requesting. In future, it will also give a reason for the decision it has made when it comes to category and priority scores.

9.4 MVP Categories

The categories:

- Malware
- Network
- Access
- Phishing
- Data Breach

The priority levels:

- Critical

- High
- Medium
- Low

Company names are automatically generated as more companies are detected within the tickets. More directories are also added as new categories and priority levels are added, allowing SecOps to easily increase the options that the AI has.

9.5 Future AI Enhancements (Semester 2)

Detailed below are the plans for future AI enhancements:

- Duplicate clustering across the entire queue (near-duplicate detection using embeddings).
- Implementation of machine learning algorithms to further expand the usability of the AI.
- Workload-aware assignment suggestions (based on engineer availability/skills).
- Trend analysis and summarisation for shift handovers.
- A confidence scoring and human feedback system to improve categorisation.
- Chatbot system to aid SecOps in alternative fixes for issues.

These will all add more utility to the AI, ensuring it is future proofed and as effective as possible for the SecOps team. This is going above and beyond the promised features from that initial meeting with Global4.

10. Backend Design (MVP on Localhost)

The backend is implemented in Python using FastAPI. For Semester 1 MVP, it runs on localhost and uses a synthetic ticket dataset. The architecture is intentionally modular to allow a controlled migration to the real AutoTask REST API in Semester 2.

10.1 Backend Responsibilities

The backend is responsible for serving ticket data to the frontend through REST endpoints, while maintaining a consistent internal Ticket schema across the system. It also integrates with the AI module to enrich each ticket with `ai.category`, `ai.score`, and `ai.reason`. In addition, the backend supports essential operations such as filtering by category, status, or priority, as well as retrieving individual tickets by ID.

10.2 Folder/Module Responsibilities (Backend)

The backend is structured into several key modules, each with a distinct responsibility.

The file `backend/data/tickets.json` contains the synthetic ticket dataset used during MVP testing. Within `backend/app/models.py`, the Pydantic Ticket model defines the stable internal schema for ticket data. Data retrieval is handled by `backend/app/providers/fake_AutoTask.py`,

which serves as the provider that reads from tickets.json. The AI logic is integrated through backend/app/services/ai_categoriser.py, which Toby will later replace or extend. Finally, backend/app/main.py defines the FastAPI routes that orchestrate the provider and AI components to produce the API responses.

10.3 Key API Endpoints (MVP)

Endpoint	Method	Purpose
/health	GET	Health check for demos and monitoring.
/api/categories	GET	Returns list of UI category buckets.
/api/tickets	GET	Returns ticket list with AI enrichment; supports filters (category/status/priority).
/api/tickets/{id}	GET	Returns a single ticket by ID with AI enrichment.

10.4 Why Localhost for MVP

We chose to use localhost for the MVP because it avoided handling real company data, removed early deployment complexity while the data model and AI interface were still evolving, and allowed fast iteration across the backend, AI, and frontend teams through a stable local API.

11. Migration Plan: Fake Provider → Real AutoTask REST API

11.1 Migration Principles

- AutoTask remains read-only; no writing back to production tickets in MVP.
- Only the provider layer changes when moving from fake data to live AutoTask API.
- Internal Ticket model remains stable to avoid refactoring AI and frontend.

11.2 What Will Change

The migration from the fake provider to the real provider will affect the structure of the backend in the following way:

- Replace the fake AutoTaskProvider with the real AutoTaskProvider which calls AutoTask REST endpoints.
- Add authentication handling for AutoTask (OAuth/client credentials, depending on tenant configuration).
- Add field mapping from AutoTask response JSON to internal Ticket model fields.

11.3 What Will Not Change

In order to maintain the software's functionality and ensure it fulfils the requirements, the following have been outlined as unchanged:

- FastAPI routes and response shapes (frontend contract).
- Ticket model used internally (unless AutoTask reveals genuinely missing fields).
- AI module interface: `categorise_ticket(title, description) → {category, score, reason}`.
- Frontend category pages and ticket display logic.

11.4 Data Storage of AI Results

AI-generated fields should be stored in the tool's own data store (planned Semester 2), not in AutoTask. This avoids reliance on AutoTask customisation limits and ensures the intelligence layer can evolve without affecting the source ticketing system.

12. Security, Privacy, and Data Handling

The MVP uses synthetic tickets only: companies (Company 1..N) and fake engineer names.

No credentials or secrets stored in source control; secrets will be handled via environment variables in Semester 2 deployment.

AutoTask remains the source of truth; the tool consumes ticket data and stores only required enrichment fields internally.

Access control and auditing are the plans for Semester 2 (authentication + role-based authorisation).

13. Testing and Validation Plan

13.1 MVP Testing

- **API smoke tests:** `/health`, `/api/categories`, `/api/tickets`, `/api/tickets/{id}`, to verify that all the data is coming through the API and being displayed properly.
- **Functional tests:** filter the tickets by category, status and priority, and validate correct outputs.
- **AI sanity tests:** ensure that there are the score and category on all tickets and validate them against known Datto RMM-style wording.
- **UI integration tests:** confirm the frontend renders the categories and ticket lists correctly from the API. This shows that everything is working correctly and in tandem with one another.

13.2 Acceptance Criteria (MVP)

- A user can open the dashboard and see the tickets grouped by category.
- Ticket lists display priority grouping and key attributes at a glance.
- Clicking a ticket shows the full attributes and description content.
- AI output exists for all tickets and is easily understood.
- System runs reliably on localhost.

14. Project Timeline and Sprint Plan

The following timeline is based on the provided Gantt chart and sprint planning document.

Phase	Dates	Focus / Outputs
Research Phase	24/11/2025 – 30/11/2025	Research AutoTask API/system connections, AI research, UI wireframes.
Prototype Development	30/11/2025 – 15/12/2025	Build MVP: Toby AI, backend API, frontend dashboard; synthetic dataset.
Programming Completion	15/12/2025 – 29/12/2025	Complete core programming for backend/frontend/AI; stabilise integration.
Final Deliverables	29/12/2025 – 04/01/2026	Final video + reflection docs; submission deadline 04/01.
Q&A Preparation & Break	04/01/2026 – 16/01/2026	Prepare Q&A session; consolidate learning and demo readiness.

See 18.6 Appendices for the Gantt chart.

Due to experiencing delays in receiving the database schema structure from Global 4 internals, our timeline had to be postponed, as the database schema structure was an integral part of ticket generation; without this the development on AI nor frontend could proceed. Furthermore, programming the AI provided slight technical challenges which took some time to resolve, as well as environment and dependency issues regarding front end frameworks; this also caused the timeline to be delayed. Regardless of this, our group successfully completed all the deliverables for the MVP before the deadline, by assisting each other wherever possible.

15. Monitoring, Evaluation & Quality Control

The project lead requested group meetings on a regular basis where possible, in which members updated each other on their progress, any setbacks currently facing and how they could be resolved. Any queries regarding the project requirement or deliverables would also be discussed and answered within these meetings (see 18.5 Appendices for minutes).

16. Future DevSecOps Deployment Goals (Semester 2 – SMART)

As mentioned previously, the project will evolve from a localhost MVP to a deployed service that can be accessed by the SecOps team and integrates with live AutoTask tickets.

16.1 SMART Goals

1. **SPECIFIC:** Deploy the backend API as a hosted service (internal VM or cloud) and the frontend as a web dashboard accessible to SecOps.
2. **MEASURABLE:** Automated build + deployment pipeline completes within 5 minutes; automated tests run on every pull request.
3. **ACHIEVABLE:** Use GitHub Actions for CI, containerisation where appropriate, and managed hosting for database.
4. **RELEVANT:** Supports secure, consistent operational use by multiple analysts and reduces triage time.
5. **TIME-BOUND:** Complete initial deployment and secure access controls during Semester 2, with a working pilot by mid-semester.

16.2 Deployment Strategy (Target)

- CI: On every pull request, it will go through code linting, unit testing and security checks, scanning for vulnerable dependencies and other issues.
- CD: Changes merged into the main branch in GitHub are automatically deployed into a staging environment to be validated. Production releases are triggered by tagged versions or merges to the prod branch.
- Secrets: All sensitive credentials and API keys are managed through secure storage solutions like GitHub Secrets.
- Monitoring: This will be done through basic health checks and request logging, with plans to add advanced metrics and more proactive alerts.

17. Risks and Mitigations

Risk	Impact	Mitigation
Delayed access to live AutoTask credentials/data	Blocks real integration testing	Use synthetic tickets + mock server; build real AutoTaskProvider later with mapping plan.
AI misclassification reduces trust	Analysts may ignore AI output	Provide explainability (ai.reason), allow human override, iterate with feedback.
Scope creep (too many features)	MVP not delivered on time	Strict MVP scope; backlog future features for Semester 2.
Security concerns with storing sensitive data	Compliance and operational risk	MVP uses only synthetic data; future storage follows the least privilege and secure secrets handling.
Team integration delays	Frontend/AI/backend not aligned	Maintain stable API contract and AI output contract; weekly integration checkpoints.

18. Appendices

18.1 Client Meeting Minutes (meeting 2):

Attendance - Availability

- Tanis, lead developer, is on leave for 3 weeks.
- Toby was unable to join the meeting due to illness.

Meeting Objectives

- Global4 asked what information we need from them to proceed.
- We confirmed the project must be future-proof and aligned with the programming languages currently in use.

Technical Preferences - Flexibility

- Front end: React.js suggested for UX.
- Back end: TypeScript and JavaScript — no strong preference from Global4.
 - We mentioned we're likely to lean more towards TypeScript due to the nature of JavaScript.
- Global4 uses custom in-house platforms but has no restrictions — we're free to use other technologies.

Data - Error Handling

- We proposed obtaining the database structure and creating bogus data instead of using real client data, ensuring accuracy.
- We requested details on the types of errors they receive.
- Issues typically come in via phone or email — two key points for creating linkage.
- We could intercept at the point of email/phone intake or via triage.
- Plan to build a separate SecOps area and dashboard, with AI programmed to pick up alerts and organise them into different sections.
- Solution must be future-proof and long-lasting.

Sample Data Requirements

- Global4 said they can provide a sample data structure which would include the fields/headers of data, such as:
 - Company

- Contact number
- Email
- Type of issue.
- Miscellaneous
- The challenge lies in the volume and variety of issues.
- Most issues come in via email or API.
- Global4 said they can give us a spread of 30–40 types of issues to build our own dummy data.
- Emails vary in size; inconsistency is the main challenge.
- Pre-triaging would help sort this out.

Ticket Data Structure

- Structure of ticket data is crucial.
- Global4 can provide:
 - Export of tickets in CSV format.
 - Sample data from a few tickets to give insight into each field.
 - Actual field names and a human-readable format.
- Their database follows a normal relational structure.

Legal - Security

- Claire mentioned they are still waiting for HR to send the company-made NDA.
- Global4 asked how we will store data and who will have access.
- We confirmed we will use GitHub for:
 - Access control
 - Version control
 - Two-factor authentication
- Assured them it is quite secure.

Project Brief - Timeline

- Claire sent us a sample project brief that she received from JJ.
 - We are to update it with our actual project details and send it back to Claire.
- Regarding sample data:
 - No strict deadline for when to receive the data regarding tickets (discussed above) from Global 4.
 - We understand their schedule is busy.
 - We can notify them along the way if we encounter any issues.

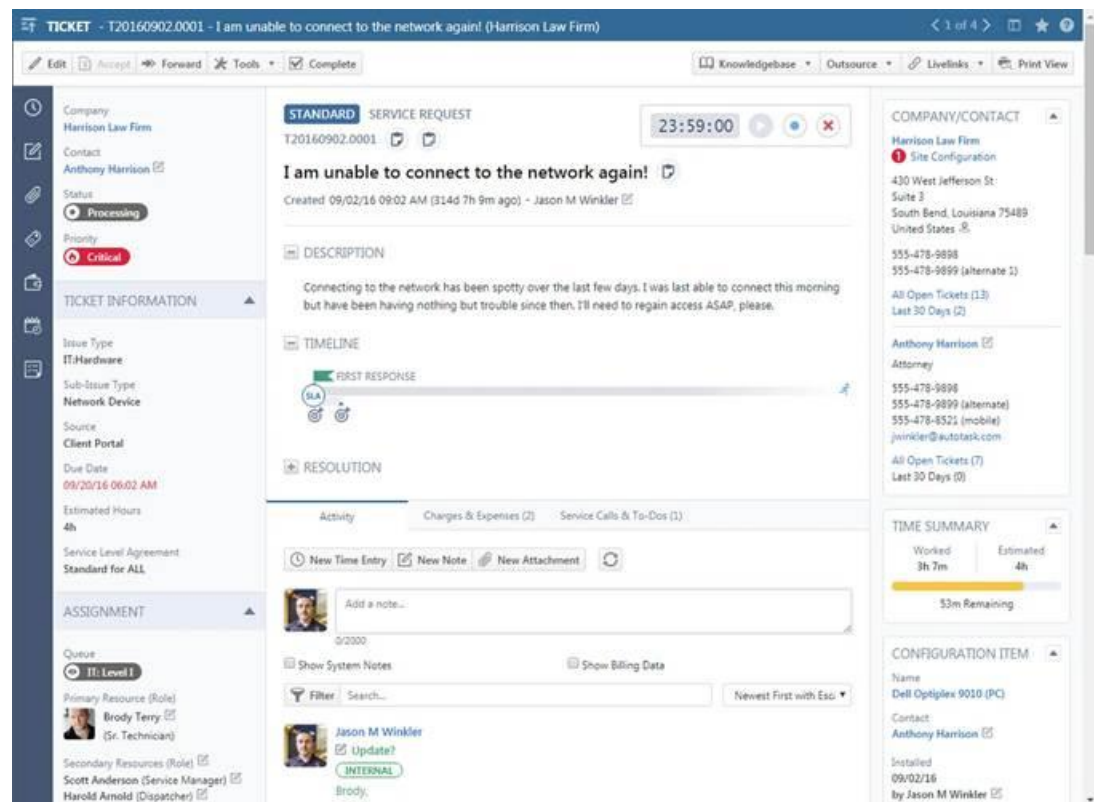
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18.2 Research on UX Principles

Important principles to adhere to and how I plan to do that:

1. **User-Centricity:** putting users' needs first and making decisions based on what you know about them and what they want from the product:
 - a. Users are SecOps
 - b. SecOps wants an interface that is easy to use and doesn't differ immensely from AutoTask as they are all familiar with it and don't want to have to be taught a new system
 - c. My idea is to implement categorisation and organisation by displaying 'cards' and 'lists' inspired by Trello, wherein each list is an AI defined category and each ticket is a 'card'.
 - d. Clicking on a ticket will expand to an almost same looking interface that SecOps see – but removing or not prioritising the less important information about a ticket:

e.



2. **Consistency:** keeping designs consistent in terms of how they look and function across all pages and screens, keeping consistency across visuals (buttons and links) and your tone:
 - a. I plan to implement a dashboard/homepage which will be the 1st page the site opens on. The homepage will display 'quick' and urgent information, quick stats about work progress.
 - b. A table will be featured in the middle of the page which will show all the tickets marked with HIGH Priority that are due that day.
 - c. To keep consistency across the pages, I can:
 - i. integrate soft rounded corners as opposed to the 2D and sharp rectangles,
 - ii. Make sections about the ticket on the side panel of a ticket appear like lists from the previous page, which each attribute mimicking the appearance of a card.
3. **Hierarchy:** relates to both information architecture and visual architecture. InfoArch– the sitemap, overall structure and organisation of the website and how the use will navigate from one page to another. VisualArch– how elements are laid out on the screen:
 - a. For navigation, I plan to implement a navbar to the side (1st design was on the left, 2nd design was on the right) which a home icon for the homepage, and icons for all active tickets and closed tickets, as well as a user icon for the account at the bottom.
 - b. I also plan to show the Breadcrumb at the top of a page underneath the company name and logo header, so it is easy to see where you came from.
 - c. For VisualArch, I plan to show tickets needing immediate attention with a read header for urgency and to draw attention, placed in the middle as it is usually where the eyes go first.
 - d. Different font sizes for headings and subheadings.
 - e. Again, putting less needed/less regarded by SecOps information towards the bottom.
 - f. Only showing the immediate information about a ticket needed at hand on the 'card' of a ticket, the rest will be shown upon clicking, which takes you to a new page.
4. **Context:** understanding who the users are and in what context they will be using the website, what factors may impact user experience:
 - a. SecOps are a small team but very busy, very swamped as they receive a large load of tickets

- b. Tickets have priorities set to them as well as due dates, meaning they usually have to be working a lot and very quickly
 - c. Therefore, it is important to show them only what is necessary so they have the knowledge they need at first glance.
 - d. Hence, I re-affirm my ideas on InfoArch & VisualArch.
5. **User Control:** giving the user the right amount of control over how they interact with a product, as they often perform unwanted actions and need a 'clearly marked emergency exit'. **NOTE: This is especially true with AI automation, and there needs to be a balance between AI and user control. Users should be able to override AI decisions.**
- a. Considering we have AI sorting and categorising the tickets, it's possible it makes a wrong decision.
 - b. I plan to put an ellipsis menu for each ticket in the categorised lists in case SecOps notices something wrong, they can assign a new category which will be done via modal.
6. **Accessibility:** ensuring the website is accessible to and usable for as many people as possible, catering to the needs of people with disabilities.
- a. I plan to use clear and plain language throughout the site
 - b. Implement a clear and semantic HTML design
7. **Usability:** learnability, efficiency, memorability, errors, satisfaction:
- a. Providing quick navigation and all the necessary information.

Reference list

Eye-Able. (2025). *WCAG 2025 simply explained: Web Content Accessibility Guidelines* / Eye-Able®. [online] Available at:

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END OF APPENDIX 18.2

18.3 Research on AI Integration Benefits

Benefits of AI:

- **Faster response/solution times**
 - Can auto-classify, route and acknowledge tickets, cutting down human interaction, and could therefore be used to auto assign tickets to a to-do list
 - Self-service/bot-deflection (knowledge base suggestions, FAQ bots) reduce load and could speed up the resolution of simple/trivial tickets
- **Improved efficiency and productivity**
 - By automating the repetitive work, SecOps can focus on the complex and high-value tickets.
 - Studies and tools show that automating portions of ticket flows can yield returns - BoldDesk says AI help desks can deliver a ROI of 100%-300% dependent on volume and how many interactions can be automated
- **Better prioritisation and quality**
 - AI can better assess the urgency of a ticket via multiple inputs - the severity, number of affected users, customer importance, SLA (service-level agreements, and similarity to past incidents. This allows for a more accurate ranking of tickets
 - Sentiment analysis and text analysis could help detect frustrations or crises that need escalating
- **Scalability and handling peak loads**
 - During load spikes (e.g. total software outages, high season, campaigns), the AI can help to reduce backlog explosion.
 - Since the AI tool can work 24/7, it could enable support functions to operate outside of business hours or in other time zones more cheaply.
- **Consistency, reduced errors and better analytics.**
 - AI reduces classification or routing errors (human mistakes, mis-tags).
 - Having structured metadata and model feedback can provide insights into ticket trends, root causes, and common failure types. You can then use that to improve product reliability, show the fault in a product and therefore reduce incoming tickets
- **Improved customer satisfaction.**
 - Faster replies, fewer back-and-forths, consistency in who they're communicating with, being routed to the correct specialist all will contribute to a better service.

- **Better SecOps experience and retention**
 - Removing tedious tasks helps to avoid burnout and lets the team do what they do best.
 - Tools that assist with summarisation and suggested responses can reduce cognitive load.

Case studies:

- **Everise:**
 - Using Retell AI, they 'contained' (resolved/prevented escalation of) 65% of internal service desk tickets with the bot, translating into roughly 600 man-hours saved per month.
- Academic examples
 - TaDaa, a real time ticket assignment bot, achieved ~95.2% accuracy on suggesting where a ticket should be sent, and ~79% accuracy top-5 accuracy for recommending the correct resolver.
 - ESSMArT (Brightsquid) cut down about 9.2 minutes per user request on average by automating summarisation, escalation prediction, etc.

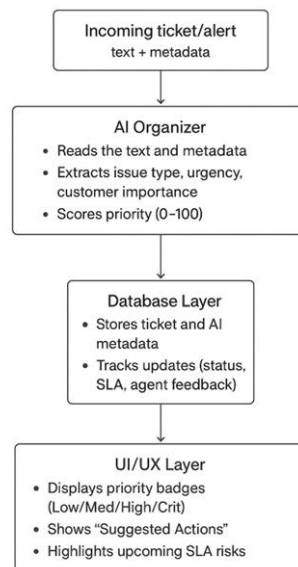
Risks and Challenges

- **Quality and accuracy issues**
 - If AI mis-classifies, mis-routes or gives inappropriate suggestions, this could lead to slow resolution, so will require human oversight.
- **Context and domain expertise**
 - It may struggle with ambiguous/domain-specific tickets, so training data must reflect real tickets, SOPs, customer types.
- **User trust and transparency**
 - Customers and SecOps may not trust the automated decisions - it should explain why and also have easy overrides.
- **Data security**
 - Tickets may have sensitive data. It must comply with GDPR, HIPAA etc, mask PII, secure storage
- **Edge cases**
 - Highly urgent or complex issues, emotional or escalated customers, etc., might need human interaction or special pathways.

Flow:

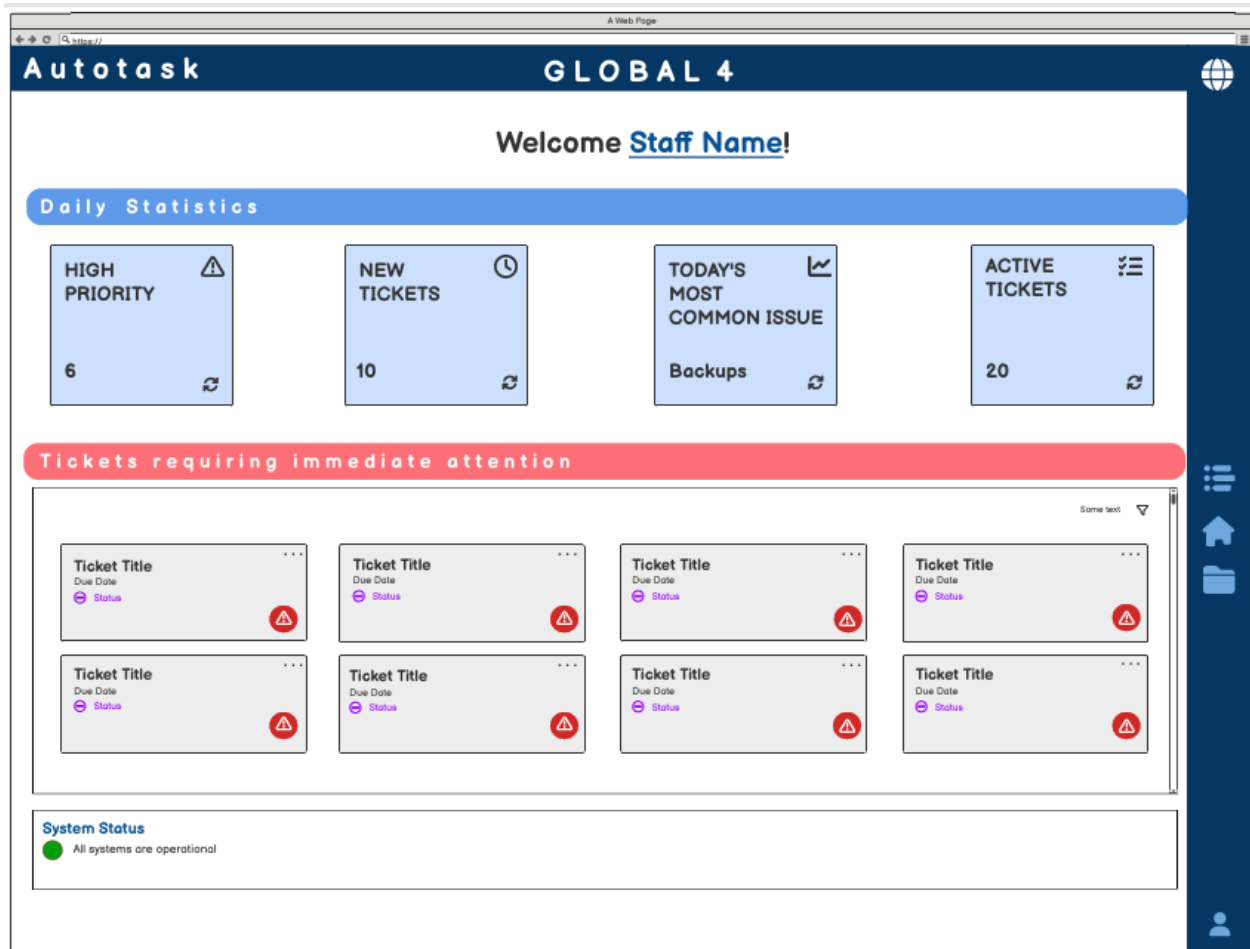
- **Incoming ticket/alert**
- **AI organiser**
 - Reads the text and metadata
 - Extracts issue type, urgency, customer importance
 - Scores priority (0-100)

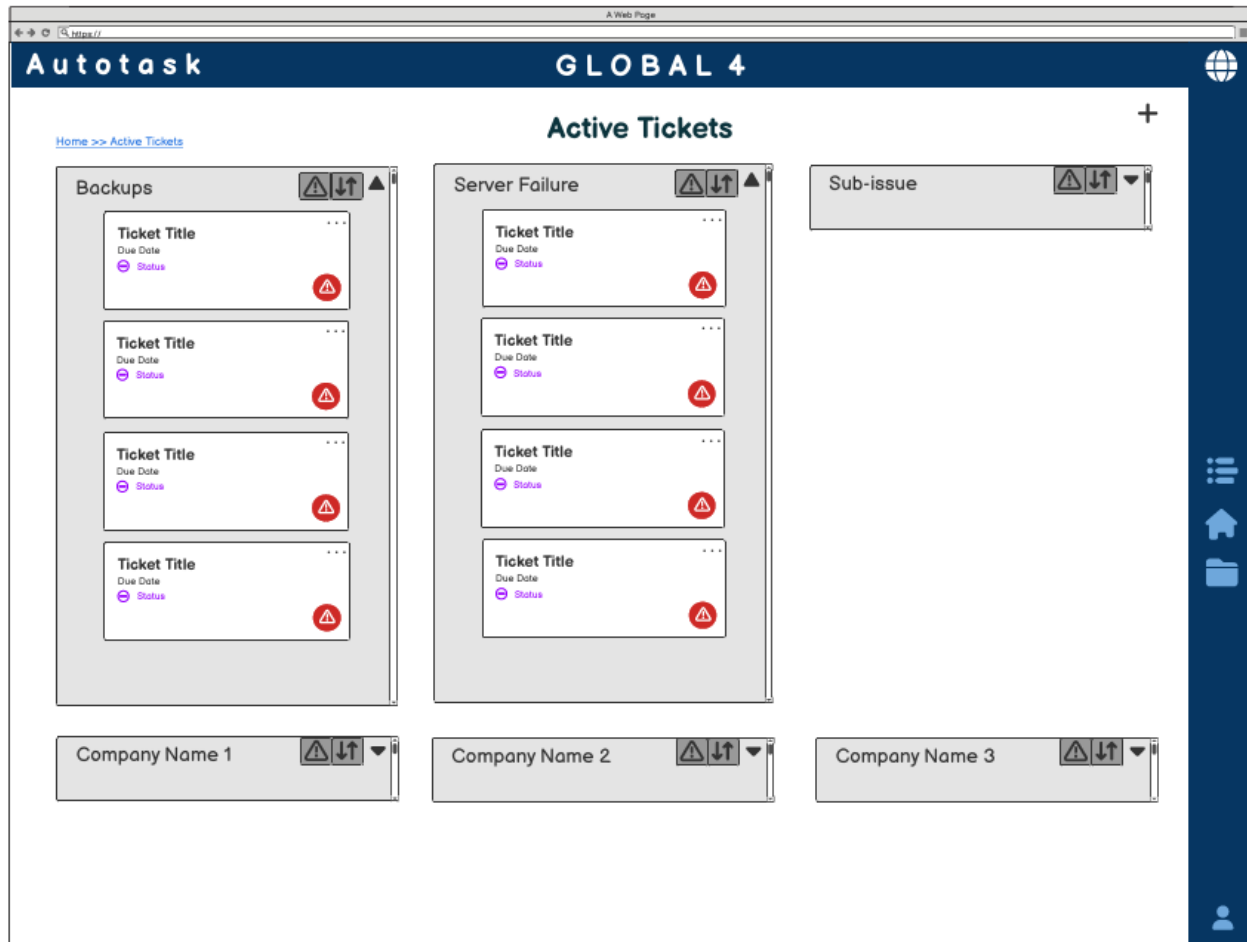
- Suggests category and assignee
- **Database Layer**
 - Stores ticket and AI metadata
 - Tracks updates (status, SLA, agent feedback)
- **UI/UX Layer**
 - Displays priority badges (Low/Med/High/Crit)
 - Shows “Suggested Actions”
 - Highlights upcoming SLA risks.

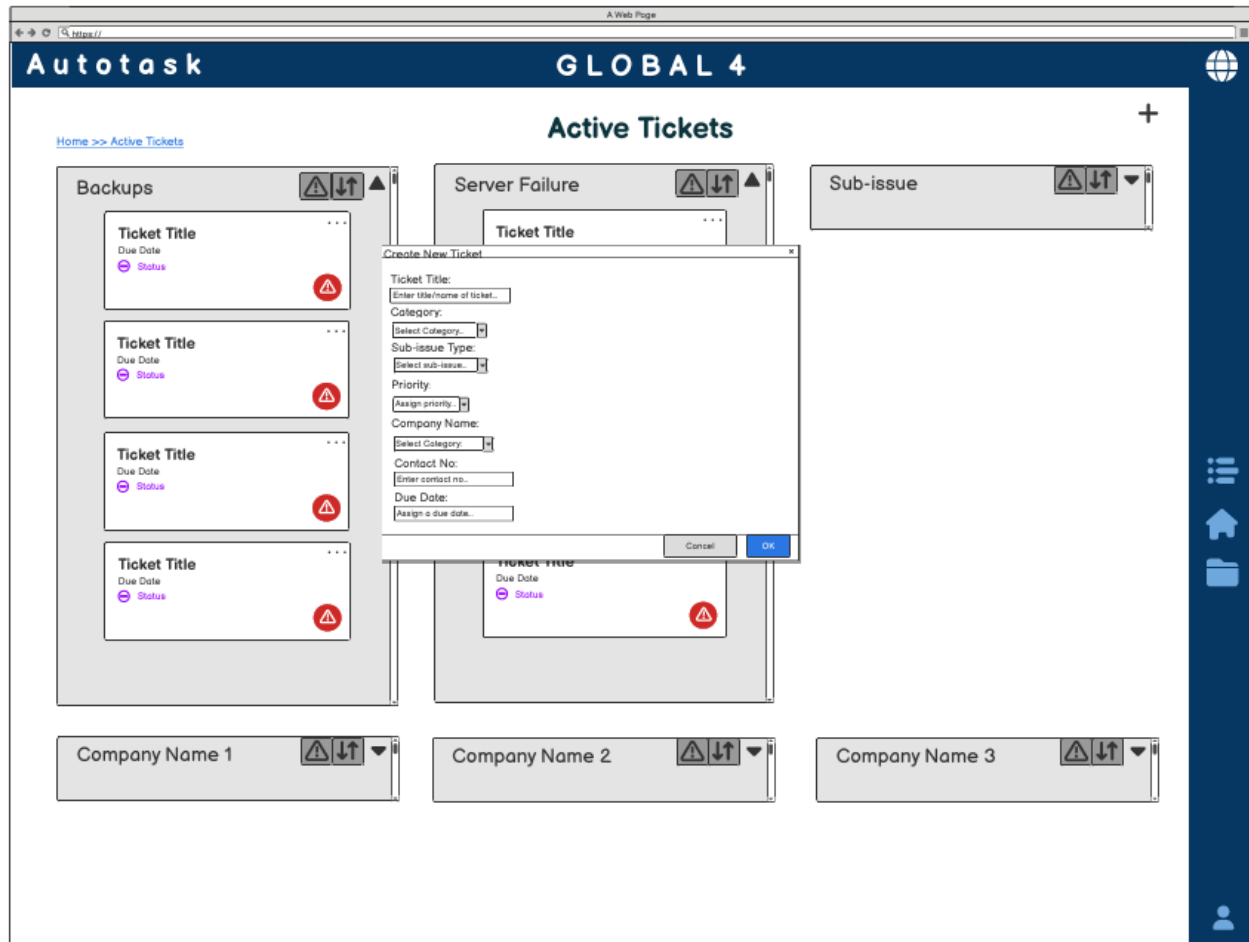


END OF APPENDIX 18.3

18.4 UX Wireframes







A Web Page

Autotask GLOBAL 4

Home >> Active Tickets >> Ticket Title

TICKET TITLE

Ticket Description

GENERAL

Company Name:

Contact No:

Tickets Status: ⊖ Status

Contact No:

Priority:

TICKET INFO

Additional Contact:

Issue Type:

Sub-issue Type:

Source:

Due Date:

ASSIGNMENT

Primary Resource:

Secondary Resource:

Co-managed Visibility:

CLOSING

Resolution:

Closer:

Closure Code:

Date Closed:

PROGRESS UPDATES

[Staff Member Name] DD/MM/YY: HH:MM

[Redacted Progress Update]

The screenshot shows the Autotask GLOBAL 4 web interface. The left sidebar contains navigation links for Home, Active Tickets, and Ticket Title. The main content area is divided into several sections:

- GENERAL**: Fields for Company Name, Contact No., Tickets Status (with a Status button), and Priority.
- TICKET INFO**: Fields for Additional Contact, Issue Type, Sub-issue Type, Source, and Due Date (highlighted in red).
- ASSIGNMENT**: Fields for Primary Resource, Secondary Resource, and Co-managed Visibility.
- CLOSING**: Fields for Resolution, Closer, Closure Code, and Date Closed.
- PROGRESS UPDATES**: A section for logging progress, featuring a "Log progress" dialog box with a text area and "Cancel" and "Save" buttons. Below this is a table for progress updates with columns for Staff Member Name and DD/MM/YYYY : HH:MM.

END OF APPENDIX 18.4

18.5 Group Meeting Minutes

Date: 11.12.25

Attendance

- All three team members attended via Teams.
- I (Khyati) attended from home since I have gone back home for Christmas

Meeting Discussion Points:

Project Hosting & Compliance

- Rearranging the way AI organisation will be hosted — functionality remains the same, but Barbara (Global4) raised hosting issues.
- Barbara will likely ask compliance-related questions.

Global 4 Operations & Resourcing

- Liam noted that operations at Global4 are not doing well; SecOps is severely understaffed.
- Role changes may be necessary to adapt to the situation.
- Liam said we may need to work this sprint without ticket data from Global4, so he could assist Toby or me.

AI Development Status

- Toby confirmed the AI is at minimum functionality and does not yet include ML.
- Toby said he can start focusing on converting data to JSON form and files.
- Toby and I discussed the format of organised tickets to ensure I can create functions that can be performed for:
 - Counting the number of tickets.
 - Categorising tickets.
- Which will help in populating lists and sections.

Front End & Team Availability

- Both Liam and Toby are willing to help with front end development while I (Khyati) am away with family in London until Tuesday, 16th December.
- I shared my progress so far, outlined next steps, and confirmed all my work has been committed and pushed to my feature branch on GitHub.

Data & Schema Requirements

- Discussed the importance of making it clear to Global4 that we **need the schema** for smooth development across all stages.

Client & Stakeholder Updates

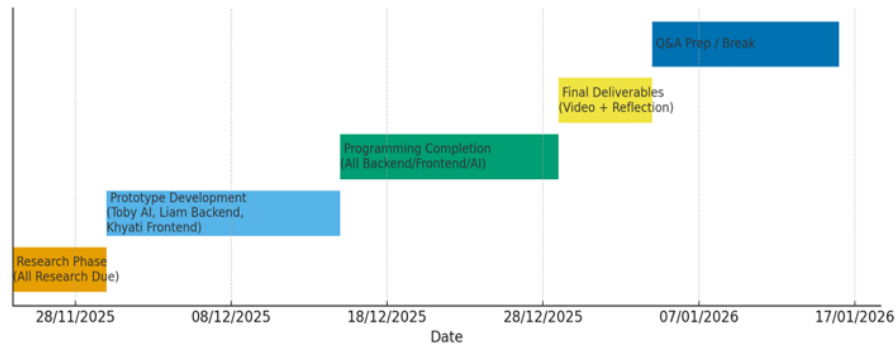
- Liam spoke to Jonathan (Claire's replacement), who is more IT-focused.
- Jonathan gave Liam permission to review project-related materials (e.g., schema) during work hours.
- Jonathan also asked Liam about placements and to discuss this matter soon.
- Liam said Global4 may want people for maintenance and expansion of this product/solution.

Scheduling

- I mentioned that the client meeting works better for me on 16th December after I return.
- If necessary, I can attend on the 15th from the hotel with my laptop.
- Liam said it's not a big issue if the meeting takes place without me.

END OF APPENDIX 18.5

18.6 Gantt Chart



Research Phase (24/11/2025 – 30/11/2025)

All research must be completed by this date. Toby is responsible for AI research, Khyati focuses on wireframes, and Liam completes backend research involving the Autotask API and system connections.

Prototype Development (30/11/2025 – 15/12/2025)

Toby works on AI programming for the prototype. Liam continues backend development, and Khyati continues frontend development. After 15 December, Toby will transition into backend, frontend, or a hybrid role to balance workload.

Programming Completion (15/12/2025 – 29/12/2025)

All programming tasks—including backend, frontend, and AI—must be fully completed by 29 December.

Final Deliverables (29/12/2025 – 04/01/2026)

During this period, the team will complete the final video presentation and personal reflection documents. The official submission deadline is 4 January.

Q&A; Preparation & Break (04/01/2026 – 16/01/2026)

After submission, the team has a break before preparing for the Q&A; session on 16 January, where questions about the software and development process will be answered.

END OF APPENDIX 18.6

END OF APPENDICES

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